

Gregory O. Lapit

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EDUCATION

Harvard University

A.B. in Physics, with Secondary in Computer Science

Cambridge, MA

June 2024

University of Alabama

M.A. in Communications

Tuscaloosa, AL

June 2026

WORK EXPERIENCE, PROJECTS & PRESENTATIONS

Advanced Computational Materials Engineering Laboratory

Tuscaloosa, AL

Material Simulation Software Researcher

September 2024 - Present

- Authored CS publication paper quantifying software performance improvements across open-source FEPX software versions. Demonstrating explicit advantages in accessibility and scalability for solving complex differential deformation problems
- Coded in Python and Bash on Linux OS to develop quality-assurance testing suites for multithreaded material simulation software. Scheduled jobs on HPC cluster daily using Slurm
- Automated validation of HDF5 data outputs from finite element analysis software for concurrent computing jobs. Accounted for parallel file system outputs using hierarchical data file format
- Designed a pipeline that included large-scale execution of scripts, post-processing of simulation data, and visualization of multi-file datasets for later publication in scientific journals

Solitude – <https://github.com/greglapit/summer-guide>

Tuscaloosa, AL

Godot Solo Game Developer

August 2025– Present

- Designed a game centered on player combat systems and a deck of cards, wielding the diamond (shard) suit as weapons. Each rank of diamond is a different weapon. Intended for publication
- Utilized Git version control daily to manage the codebase's iterative development and feature integration
- Architected inter-system communication to decouple gameplay components, enabling modular development and centralized state management
- Authored GPU shader programs using GLSL (OpenGL) programmable graphics pipeline using vertex transformations and fragment processing. Using placeholder art until commissions near publication
- Implemented modular gameplays loop with a strong focus on player interaction systems: climbing through ranks of ramping weapons (e.g., daggers, shields, scythes, staves), interacting with shops and NPCs, and unlocking narrative components. Emphasized player agency and scalable mechanics

Hoffman Laboratory

Cambridge, MA

Semiconductor & Thin Film Materials Researcher

June 2022 - May 2024

- Presented a novel precursor inlet deposition method; presented findings at the American Physical Society (APS) conference (March 2023)
- Fabricated semiconductor thin films using molecular beam epitaxy (MBE) daily; working knowledge of chemical vapor deposition (CVD) and self-constructed ultra-high vacuum (UHV)
- Performed material characterization through nanoscale analysis: Atomic Force Microscopy (AFM), Scanning Electron Microscopy (SEM), X-ray photoelectron spectroscopy (XPS)
- Designed and executed controlled growth experiments, producing detailed documentation of microscopic features for failure analysis and technical presentations; iteratively optimized deposition parameters
- Configured and integrated growth-system hardware, including PID controllers, linear manipulators, chamber ports, effusion cells, infrared temperature sensors, and ion gauges

Boeing – Maintenance Ground Operations Systems

Seal Beach, CA

Aerospace Support Equipment Design Engineer

June 2023 - September 2023

- Modeled and maintained CAD designs for Boeing 737 ground support and maintenance equipment. Documented updates and contributed to Boeing documentation daily
- Designed prototype tooling to support the installation and removal of landing gear fasteners
- Authored and updated engineering designs for patented Boeing aircraft maintenance tools

ADDITIONAL ACTIVITIES

DI Varsity Track and Field

Tuscaloosa, AL

University of Alabama and Harvard Long Sprinter

August 2019 – June 2026

- NCAA Nationals Qualifier, five-time Ivy-League Champion
- Harvard school record holder in two events, Ivy League record holder, New England record holder